12TH BIOLOGY · CHAPTER 18 · FULL CHAPTER MCQS

Inheritance

76 topic-wise MCQs from the complete chapter with answers and reasons

18.1 – 18.2 MENDEL'S LAWS AND PROBABILITY

1. The first scientist to explain the mechanism of inheritance was:

(a) Morgan (b) Sutton
(c) Mendel (d) Correns

Answer: (c) Mendel — Gregor Johann Mendel was an Austrian monk.

2. The F_2 phenotypic ratio of a monohybrid cross is:

(a) 1:1 (b) 3:1
(c) 9:3:3:1 (d) 1:2:1

Answer: (b) 3:1 — 3 dominant : 1 recessive.

3. The 3:1 ratio of F_2 is really a disguised:

(a) 1:1 ratio (b) 2:1 ratio
(c) 9:3:3:1 ratio (d) 1:2:1 ratio

Answer: (d) 1:2:1 ratio — 1 pure dominant : 2 hybrid : 1 pure recessive.

4. Alternative forms of a gene are called:

(a) Alleles (b) Loci
(c) Gametes (d) Polygenes

Answer: (a) Alleles — They lead to alternative traits.

5. The location of a gene on a chromosome is its:

(a) Allele (b) Karyotype
(c) Genotype (d) Locus

Answer: (d) Locus — Each gene has a particular locus.

6. An individual with two different alleles of a gene is:

(a) Homozygous (b) Haploid
(c) Heterozygous (d) True-breeding

Answer: (c) Heterozygous — Example: Rr.

7. The observable character of an individual is its:

(a) Phenotype (b) Genotype
(c) Locus (d) Allele

Answer: (a) Phenotype — The genotype is the blueprint.

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8. A testcross involves crossing the dominant phenotype with a:

(a) Homozygous dominant (b) Recessive phenotype
(c) Heterozygous (d) Dihybrid

Answer: (b) Recessive phenotype — It reveals whether the dominant individual is RR or Rr.

9. If a round-seeded plant is Rr, its testcross gives round and wrinkled in the ratio:

(a) 3:1 (b) 1:2:1
(c) All round (d) 1:1

Answer: (d) 1:1 — Rr × rr gives 1 Rr : 1 rr.

10. The genotype of Mendel's F_1 dihybrid was:

(a) RRYY (b) rryy
(c) RrYy (d) RrYY

Answer: (c) RrYy — It is heterozygous for both genes.

11. How many kinds of gametes does RrYy produce?

(a) Four (b) Two
(c) Eight (d) Sixteen

Answer: (a) Four — RY, Ry, rY and ry.

12. The F_2 phenotypic ratio of a dihybrid cross is:

(a) 3:1 (b) 9:3:3:1
(c) 1:2:1 (d) 9:3:4

Answer: (b) 9:3:3:1 — Four phenotypes appear.

13. Independent assortment occurs because homologous pairs line up randomly during:

(a) Mitosis (b) Fertilization
(c) Meiosis (d) Interphase

Answer: (c) Meiosis — This gives different allele combinations.

14. The number of possible allele combinations in a human gamete is over:

(a) 8 million (b) 46
(c) 23 (d) 64

Answer: (a) 8 million — 2^{23} combinations.

15. When one gene controls multiple characters, it is called:

(a) Epistasis (b) Codominance
(c) Linkage (d) Pleiotropy

Answer: (d) Pleiotropy — It distorts expected ratios.

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16. The probability of an aa offspring from Aa × Aa is:

(a) 1/2 (b) 1/4
(c) 3/4 (d) 1/16

Answer: (b) 1/4 — 1/2 × 1/2 by the product rule.

17. The probability of round yellow offspring from RrYy × RrYy is:

(a) 3/16 (b) 1/16
(c) 9/16 (d) 3/4

Answer: (c) 9/16 — 3/4 × 3/4.

18. The rule used for mutually exclusive events is the:

(a) Product rule (b) Landsteiner's rule
(c) Basic rule (d) Sum rule

Answer: (d) Sum rule — Individual probabilities are added.

19. The chance of getting two sixes with two dice is:

(a) 1/36 (b) 1/12
(c) 1/6 (d) 1/3

Answer: (a) 1/36 — 1/6 × 1/6.

18.3 – 18.4 DOMINANCE RELATIONS AND BLOOD GROUPS

20. Pink flowers in the four-o'clock plant are an example of:

(a) Complete dominance (b) Incomplete dominance
(c) Co-dominance (d) Epistasis

Answer: (b) Incomplete dominance — The heterozygote RW is intermediate.

21. The F_2 ratio in incomplete dominance is:

(a) 1:2:1 (b) 3:1
(c) 9:3:3:1 (d) 1:1

Answer: (a) 1:2:1 — Red : pink : white.

22. The MN blood group is an example of:

(a) Incomplete dominance (b) Epistasis
(c) Co-dominance (d) Polygenic inheritance

Answer: (c) Co-dominance — Both M and N antigens appear.

23. The MN system was discovered by:

(a) Mendel and Morgan (b) Sutton and Correns
(c) Bayliss and Starling (d) Landsteiner and Levine

Answer: (d) Landsteiner and Levine — In 1927.

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24. The ABO gene has how many alleles in the population?

(a) Three (b) Two
(c) Four (d) Six

Answer: (a) Three — I^A, I^B and i.

25. The genotype of blood group O is:

(a) I^AI^B
(b) I^Ai
(c) ii
(d) I^Bi

Answer: (c) ii — Allele i produces no antigen.

26. The number of blood group systems recognized in humans is:

(a) 4 (b) 23
(c) 64 (d) 43

Answer: (d) 43 — By the International Society of Blood Transfusion.

27. The ABO system was discovered by Landsteiner in:

(a) 1927 (b) 1901
(c) 1930 (d) 1952

Answer: (b) 1901 — He found antigen-A and antigen-B.

28. The gene I of the ABO system is located on chromosome:

(a) 9 (b) 7
(c) 19 (d) X

Answer: (a) 9 — I stands for iso-haem-agglutinogen.

29. Blood group A has which antibodies in plasma?

(a) Anti-A (b) Anti-B
(c) Both (d) None

Answer: (b) Anti-B — Antibodies are against the antigen that is absent.

30. Blood group AB plasma contains:

(a) Anti-A only (b) Anti-B only
(c) Both anti-A and anti-B (d) Neither anti-A nor anti-B

Answer: (d) Neither anti-A nor anti-B — So AB is the universal recipient of the ABO system.

31. Clumping of RBCs in an incompatible transfusion is called:

(a) Coagulation (b) Haemolysis
(c) Agglutination (d) Thrombosis

Answer: (c) Agglutination — Recipient antibodies attack donor antigens.

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32. The secretor gene “Se” is located on chromosome:

(a) 19 (b) 9
(c) 7 (d) Y

Answer: (a) 19 — Secretors release antigens in body fluids.

33. The Rh-antigen is also called:

(a) Antigen-A (b) Antigen-D
(c) Antigen-H (d) Antigen-M

Answer: (b) Antigen-D — It is also called the Rh factor.

34. The genotype of an Rh-negative person is:

(a) DD (b) Dd
(c) dd (d) DD or Dd

Answer: (c) dd — Allele d does not form the Rh-antigen.

35. The actual universal donor is:

(a) AB-positive (b) AB-negative
(c) O-positive (d) O-negative

Answer: (d) O-negative — It has no A, B or Rh antigens.

36. The actual universal recipient is:

(a) AB-positive (b) O-negative
(c) A-positive (d) AB-negative

Answer: (a) AB-positive — It has no anti-A, anti-B or anti-Rh antibodies.

37. Erythroblastosis foetalis occurs when the mother is _____ and the foetus is _____.

(a) Rh-positive, Rh-negative (b) Rh-negative, Rh-negative
(c) Rh-negative, Rh-positive (d) Rh-positive, Rh-positive

Answer: (c) Rh-negative, Rh-positive — The mother makes anti-Rh antibodies.

38. In erythroblastosis foetalis, jaundice is caused by the build-up of:

(a) Melanin (b) Bilirubin
(c) Fibrin (d) Opsin

Answer: (b) Bilirubin — It comes from the breakdown of RBCs.

39. To prevent erythroblastosis foetalis, the mother is given:

(a) Rh-antigen (b) Vitamin D
(c) Insulin (d) Rh-antiserum

Answer: (d) Rh-antiserum — It contains anti-Rh antibodies.

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18.5 – 18.6 POLYGENIC INHERITANCE, EPISTASIS AND LINKAGE

40. Polygenic traits show:

(a) Continuous variation (b) Discontinuous variation
(c) No variation (d) Only two phenotypes

Answer: (a) Continuous variation — Many genes add their effects.

41. Human skin colour is controlled by about:

(a) 3 genes (b) 20 genes
(c) One gene (d) 60 genes

Answer: (d) 60 genes — Three are taken for understanding.

42. The ratio of skin colours from AaBbCc × AaBbCc is:

(a) 9:3:3:1 (b) 1:6:15:20:15:6:1
(c) 9:3:4 (d) 12:3:1

Answer: (b) 1:6:15:20:15:6:1 — From very dark to very light.

43. The genotype of white wheat grains is:

(a) AABBCC (b) AaBbCc
(c) aabbcc (d) AAbbCC

Answer: (c) aabbcc — It produces no pigment.

44. Epistasis is an example of:

(a) Allelic interaction (b) Sex linkage
(c) Multiple alleles (d) Non-allelic interaction

Answer: (d) Non-allelic interaction — Alleles of different genes interact.

45. The allele whose effect is masked in epistasis is called:

(a) Hypostatic (b) Epistatic
(c) Dominant (d) Codominant

Answer: (a) Hypostatic — The masking allele is epistatic.

46. In Labrador, the genotype ee gives a coat that is:

(a) Black (b) Yellow
(c) Brown (d) White

Answer: (b) Yellow — Allele e blocks pigment deposition.

47. The F_2 ratio of coat colour in Labrador (BbEe × BbEe) is:

(a) 9:3:3:1 (b) 12:3:1
(c) 9:3:4 (d) 1:2:1

Answer: (c) 9:3:4 — Black : brown : yellow.

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48. The ratio of flower colour in foxglove (DdWw selfed) is:

(a) 9:3:4 (b) 12:3:1
(c) 9:7 (d) 15:1

Answer: (b) 12:3:1 — White : dark red : light red.

49. The Bombay phenotype is due to the genotype:

(a) ii (b) ee
(c) dd (d) hh

Answer: (d) hh — No H substance is made.

50. The chromosomal theory of inheritance was proposed by:

(a) Mendel (b) Landsteiner
(c) Sutton (d) Morgan

Answer: (c) Sutton — In 1902.

51. Humans have about how many genes?

(a) 20,000 (b) 13,000
(c) 46 (d) 2 million

Answer: (a) 20,000 — *Drosophila* has about 13,000.

52. Genes present on the same chromosome are called:

(a) Multiple alleles (b) Polygenes
(c) Linked genes (d) Holandric genes

Answer: (c) Linked genes — They tend to stay together.

53. If only parental combinations appear in the offspring, it shows:

(a) Independent assortment (b) Epistasis
(c) Incomplete linkage (d) Complete linkage

Answer: (d) Complete linkage — The genes are very close.

54. Morgan studied linkage of wing length and abdomen width in:

(a) Pea (b) Drosophila
(c) Grasshopper (d) Labrador

Answer: (b) Drosophila — He saw only parental combinations in F_2.

55. Crossing over takes place during:

(a) Meiosis-I (b) Mitosis
(c) Meiosis-II (d) Fertilization

Answer: (a) Meiosis-I — Homologous chromatids exchange DNA.

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18.7 – 18.8 SEX DETERMINATION AND SEX LINKAGE

56. Chromosomes that are similar in male and female are called:

(a) Autosomes (b) Sex chromosomes
(c) Homologues (d) Chromatids

Answer: (a) Autosomes — One pair differs and determines sex.

57. Humans have how many pairs of autosomes?

(a) 3 (b) 11
(c) 23 (d) 22

Answer: (d) 22 — Plus one pair of sex chromosomes.

58. In the XX-XY system, the heterogametic sex is the:

(a) Female (b) Both
(c) Male (d) Neither

Answer: (c) Male — Males make X and Y sperms.

59. A male grasshopper has how many chromosomes?

(a) 22 (b) 23
(c) 24 (d) 46

Answer: (b) 23 — 22 autosomes + one X.

60. A gamete without any sex chromosome is called a:

(a) Zygote (b) Polar body
(c) Nullo gamete (d) Hemizygous gamete

Answer: (c) Nullo gamete — It occurs in the XX–XO system.

61. In birds, the female is:

(a) ZW (b) ZZ
(c) XX (d) XO

Answer: (a) ZW — The male is ZZ.

62. The gene SRY is present on the:

(a) X chromosome (b) Chromosome 7
(c) Chromosome 9 (d) Y chromosome

Answer: (d) Y chromosome — It starts the development of testes.

63. Sex linkage was discovered in 1910 by:

(a) Mendel (b) Morgan
(c) Sutton (d) Correns

Answer: (b) Morgan — He found a white-eyed male fly.

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64. In Morgan's F_2, all white-eyed flies were:

(a) Males (b) Females
(c) Both sexes (d) Sterile

Answer: (a) Males — The gene is on the X chromosome.

65. If a father has an X-linked recessive trait, all his daughters will be:

(a) Affected (b) Free of the allele
(c) Carriers (d) Males

Answer: (c) Carriers — Daughters receive their father's X.

66. Vitamin D-resistant rickets is a/an:

(a) X-linked recessive trait (b) Autosomal trait
(c) Y-linked trait (d) X-linked dominant trait

Answer: (d) X-linked dominant trait — It occurs in equal ratio in both sexes.

67. Genes located only on the Y chromosome are called:

(a) Polygenes (b) Holandric genes
(c) Linked genes (d) Epistatic genes

Answer: (b) Holandric genes — They pass from father to son.

68. Hypertrichosis is a/an:

(a) Y-linked trait (b) X-linked trait
(c) Sex-limited trait (d) Autosomal trait

Answer: (a) Y-linked trait — Hair on the rim of the pinna.

69. Haemophilia A is due to abnormal clotting factor:

(a) XI (b) IX
(c) VIII (d) VII

Answer: (c) VIII — Haemophilia B is due to factor IX.

70. Which haemophilia is an autosomal recessive trait?

(a) Haemophilia A (b) Haemophilia B
(c) None (d) Haemophilia C

Answer: (d) Haemophilia C — It is due to factor XI.

71. The gene for blue opsin is located on:

(a) X chromosome (b) Autosome 7
(c) Y chromosome (d) Chromosome 9

Answer: (b) Autosome 7 — Red and green opsin genes are on X.

72. Red-blindness is called:

(a) Tritanopia (b) Deuteranopia
(c) Protanopia (d) Monochromacy

Answer: (c) Protanopia — Green-blindness is deuteranopia.

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73. In Duchenne muscular dystrophy, the missing muscle protein is:

(a) Opsin (b) Melanin
(c) Fibrin (d) Dystrophin

Answer: (d) Dystrophin — Muscle fibres die.

74. Milk production in female mammals is a:

(a) Sex-linked trait (b) Sex-limited trait
(c) Sex-influenced trait (d) Y-linked trait

Answer: (b) Sex-limited trait — The genes are present in both sexes.

75. Baldness in humans is a:

(a) Sex-influenced trait (b) Sex-limited trait
(c) Y-linked trait (d) X-linked dominant trait

Answer: (a) Sex-influenced trait — Its allele is dominant in men.

76. Amniocentesis is generally done during which weeks of pregnancy?

(a) 5th–8th (b) 30th–35th
(c) 15th–20th (d) After birth

Answer: (c) 15th–20th — Amniotic fluid with foetal cells is taken.

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